

Evaluating RAG Under Adversarial Data at Scale

The Inverse Scaling Law of Semantic Poisoning and the Dynamics of LLM-as-a-Judge

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- **Background:** Retrieval-Augmented Generation (RAG) significantly reduces hallucinations by grounding Large Language Models (LLMs) in retrieved documents.
- **The Vulnerability:** What happens when the retrieved documents are corrupted or semantically mutated? This creates a conflict between the LLM's parametric memory and the corrupted context.
- **Objective:** Empirically evaluate RAG systems using 416 AI-generated semantic mutations to understand defensive behaviors and scaling laws.

- **Dataset:** 416 semantic mutations altering factual claims, numbers, and code.
 - *Public Data:* FastAPI documentation.
 - *Private Data:* Internal system documentation (no pretraining exposure).
- **Models Evaluated:** Llama 3.1 8B vs. 17B.
- **Configurations:** Tested with and without defensive System Prompts.
- **Evaluation Pipeline:** LLM-as-a-Judge (Llama 4 Scout 17B), cross-validated with human reviewers ($k > 0.73$).

Evaluation Metrics (Labels)

Every response was classified into one of four mutually exclusive labels:

- ➊ **Abstain (Safe):** The model recognizes contradictions and refuses to answer.
- ➋ **Faithful (Fooled):** The model blindly repeats the poisoned context.
- ➌ **Inconsistent (Violation):** The model silently corrects the error using its internal knowledge (violating RAG's groundedness principle).
- ➍ **Hallucination:** The model fabricates completely unrelated information.

Finding 1: The Inverse Scaling Law

Hypothesis: Larger models are safer and more resistant to attacks.

Reality: Without a System Prompt, larger models are *more vulnerable*.

Model (No Prompt)	Abstain Rate (Safety)
Llama 3.1 17B	3.61%
Llama 3.1 8B	42.55%

Why? The 17B model's superior reasoning allows it to seamlessly synthesize flawed logic (Faithful: 51.4%) or silently fix it (Inconsistent: 44.7%). The 8B model gets confused and safely abstains.

Finding 2: The Parametric Shield

What happens when the model evaluates **Private Data** (no pretraining exposure)?

- The model loses its “Parametric Shield” (internal knowledge).
- **Inconsistent (self-correction)** rate plummeted to just 2.88%.
- **Faithful (fooled)** rate jumped to 45.19%.

Conclusion

Enterprise RAG systems deployed on internal/private data are significantly more vulnerable to semantic poisoning than those using public data.

Finding 3: Prompt Engineering is a Shield, Not a Cure

- Adding a defensive System Prompt effectively acts as an immune response:
 - 17B Abstain rate: 3.61% → **60.10%**
 - 8B Abstain rate: 42.55% → **63.70%**
- However, **no configuration** achieved the $\geq 90\%$ Abstain safety threshold.
- **Conclusion:** Lexical instructions alone cannot override deep semantic manipulation.

Finding 4: The Hallucination Myth

- **Hallucinations** are often considered the biggest threat in LLMs.
- Across over 2,000 automated evaluations, the hallucination rate **never exceeded 0.24%** (effectively zero).
- When poisoned, RAG models do not invent random facts. They exhibit a binary failure mode:
 - 1 Passively trust the poison (Faithful).
 - 2 Rebel and silently fix the poison (Inconsistent).
- **Shift in Focus:** Security research must pivot from measuring “hallucinations” to measuring “faithfulness to poison.”

- **Scale vs. Safety:** RAG vulnerability scales *inversely* with model size when left unprotected.
- **Data Privacy Risk:** Private, internal data exposes RAG pipelines to much higher risks due to the absence of the Parametric Shield.
- **Future Directions:** While prompt engineering helps, robust RAG systems will require structural and architectural defenses (e.g., GraphRAG, Controlled Noise Injection) to withstand semantic attacks.

Thank You!